

GIFTANATOR

FUZZY-LOGIC GIFT GENIE ENHANCING THE HUMAN ART OF GIFTING

Final presentation in Fuzzy Sets and Systems I

University of Fribourg, 20.12.2024

THE INITIAL IDEA

Overview

- The Fuzzy Giftanator is a Python-based application that personalizes gift-giving using fuzzy logic.
- Users can enjoy tailored gift suggestions or a fun guessing mini-game.
- Balance between modern technology and the human art of meaningful gifting.

Pains

- Gift selection can be stressful, such as uncertainty with preferences, relationships, and occasions.
- Technology often isolates creativity, disconnecting us from the real-world emotions and traditions of gift-giving.
- Current solutions lack a personal touch, offering overly generic suggestions.

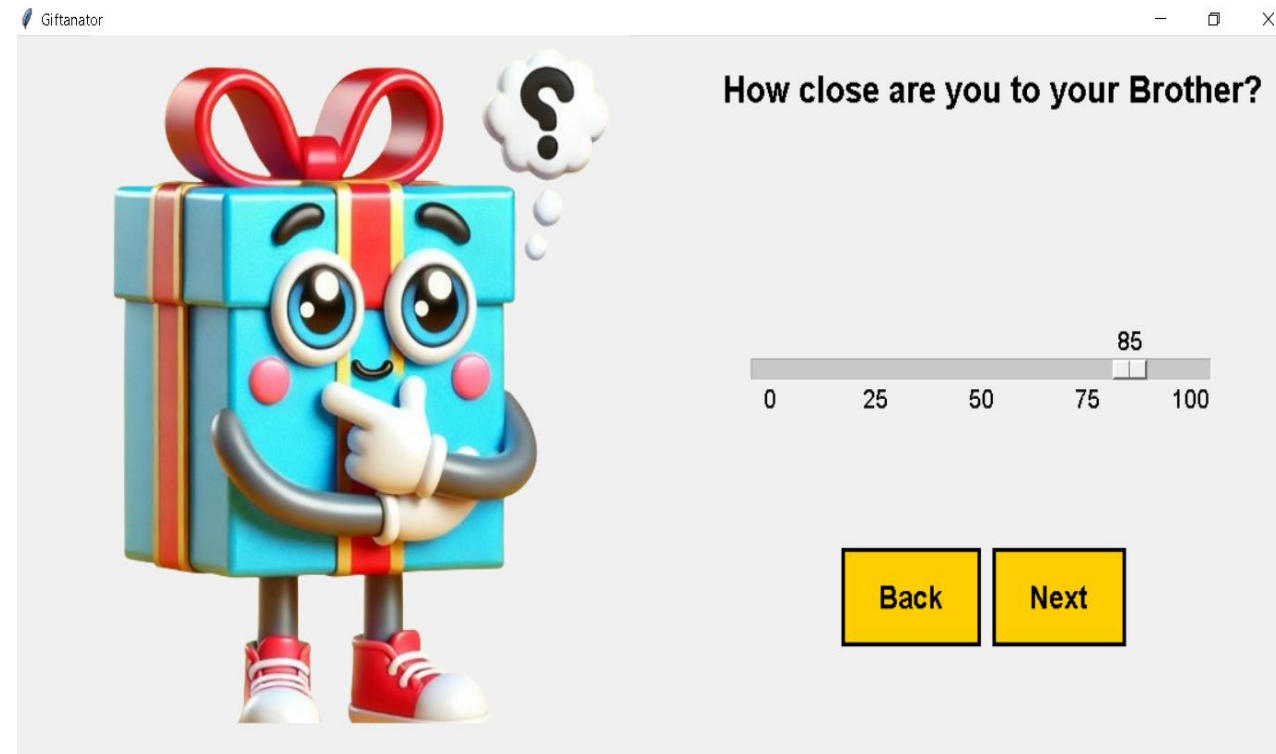
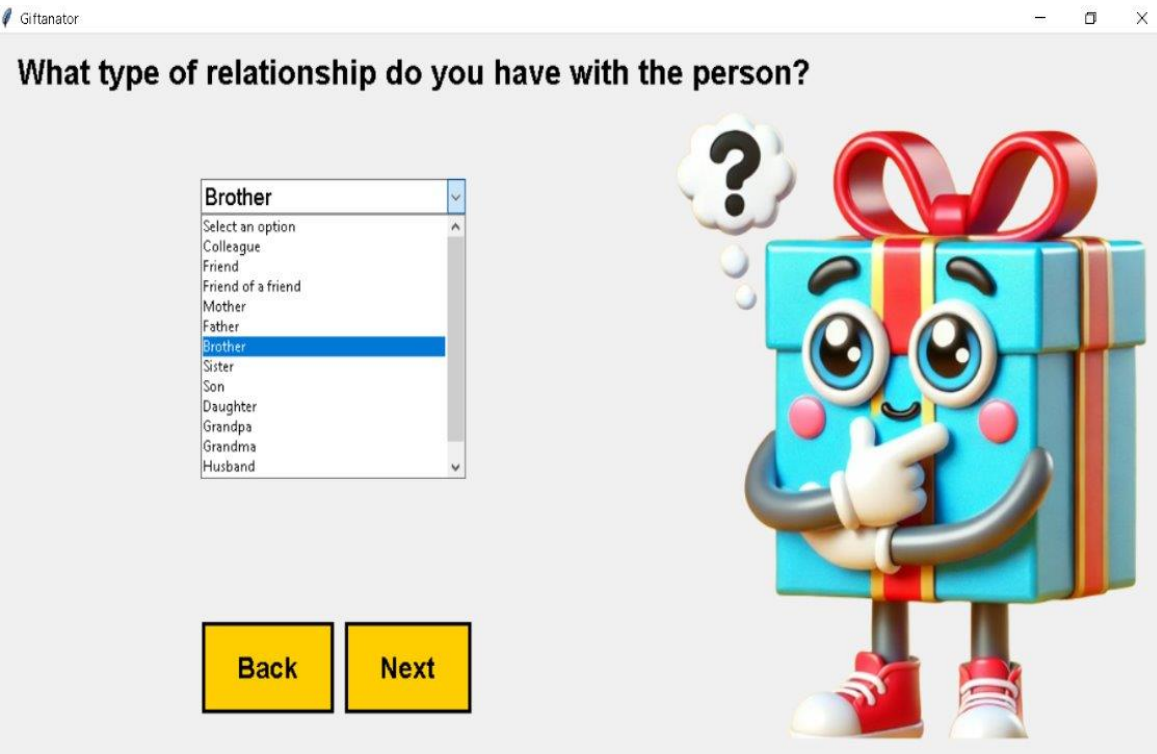




GOALS (RECAP)

- Enhanced personalization by mimicking human intuition for meaningful gifts using fuzzy logic
- Boosting the engagement with a gamification experience through an interactive gift-guessing mini-game
- Incorporate flexible decision-making to reflect real-life gifting

OUR FRONT-END



OUR BACK-END

Gift Recommender

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Questions and 'user1's' responses to them are given as below -
Question_1: What type of relationship do you have with the person?
Answer_1: Colleague
Question_2: How close are you to your Colleague?
Answer_2: 0
Question_3: What is the occasion?
Answer_3: Birthday
Question_4: How important is this Birthday to you?
Answer_4: 0
According to the answers to above questions you have to provide 3 gift suggestions for 'user2'.

The set of gifts from which you have to recommend is:
{'for_friends': {'Under_20_dollar': ['UNO Card Game', 'Personalized Coffee Mug from Etsy', 'Mini LED Keychain Projector']
```

Guess the Gift

AI host:
This is a beautiful and iconic luxury watch. It is known for its precision, durability, and timeless design. Can you think of any watch brand that matches this description?

Timex watch

User:
Timex watch

AI host:
Hmm, Timex is a great brand, but the surprise gift is actually from a higher-end luxury brand. This watch is often seen as a status symbol and is highly coveted by watch enthusiasts.

Tissot watch

User:
Tissot watch

AI host:
Tissot watches are also a great choice, but the surprise gift is from a brand that is even more renowned in the luxury watch industry. This brand is synonymous with prestige, craftsmanship, and timeless elegance.

Rolex watch

User:
Rolex watch

SWOT ANALYSIS

SURVEY WITH OTHER GROUPS

Strengths

- Personalization
- Engagement
- Fuzzy Approach
- User Feedback Integration

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Weaknesses

- Limited to web
- Potential Complexity
- Database Dependence

W

Opportunities

- Market Gap
- Partnerships
- Seasonal Trends

O

Threats

- Competition
- User Expectation
- Data Accuracy
- Technological Fatigue

T

☐ Gunakar: How do you evaluate gift suggestions?

- Feedback from users (Reinforcement Learning with Human Feedback)
- Time spent on the platform
- Return rate of users for suggestions

☐ Gobi: How do you keep up with current trends?

- Regularly update the database with the latest products (e.g., using Google search suggestions)

☐ Deep: Is a mobile app suitable for gifting, and how do you handle disliked suggestions?

- Web app preferred, since the usage is occasional
- Users can click on “re-suggest” and answer more questions for better suggestions

COMPARISON WITH 2 DIFFERENT EXISTING PLATFORMS

	GiftGen.co.uk	Fuzzy Giftanator
Personalization Approach	Provides gift ideas based on specific criteria like age and gender.	Considers a broader range of factors with varying degrees (closeness, budget flexibility).
User Interaction	Users input specific details to get suggestions.	Gathers information conversationally.
Gamification	No gamified elements.	Includes a guessing game to enhance user engagement and add fun.
Decision-Making Process	Fixed categories and crisp logic.	Flexible, fuzzy logic interprets and processes user inputs.
Budget Comparison	Users set specific budget constraints.	Allows users to input a range or degree of affordability, providing adaptable options.

	Geschenkidee.ch	Fuzzy Giftanator
Personalization Approach	Offers categorized gift ideas by occasion and recipient.	Tailors suggestions based on nuanced inputs like relationship depth and occasion significance.
User Interaction	Users browse predefined categories.	Users interact via Q&A for refined suggestions.
Gamification	No gamified elements.	Includes a guessing game to enhance user engagement and add fun.
Decision-Making Process	Traditional e-commerce approach with crisp logic.	Fuzzy logic accommodates complexities of human preferences.
Budget Comparison	Users select predefined price ranges for gifts.	Allows users to input a range or degree of affordability, offering more flexible suggestions.

EXAMPLES OF CRISPY SOLUTIONS



Gift Ideas Generator

1	Spend	2	Age	3	Personality	4	Gender
	Nothing		0 - 1		Sporty / Fit		Male
	€3		1 - 2		Creative / Arty		Female
	€5		3 - 4		Nature Lover		Either
	€10		5 - 7		Lively / Bubbly		Couple
	€15		8 - 9		Fashionable		
	€20		10 - 12		Gadget / PC Mad		
	€30		13 - 15		Hobbyist		
	€40		16 - 17		DIY Enthusiast		
	€60		18 - 30		House Proud		
	€90		31 - 40		Intellectual		
	€150		41 - 55		Adventurous		
	€250		56 - 64		Hippy / Spiritual		
	€400		65 - 75		Indoor Type		
	€600+		75+		Outdoor Type		

Search



Gift Ideas around €60 for 18-30 year old Hippy / Spiritual, Adventurous people.

Like 89

	Websites	Price Compare	Shops
Adventure Day Experience	buyagift.co.uk	ShopWiki	Argos
Bicycle Lights	amazon.co.uk	ShopWiki	Halfords
Falconry Experience	buyagift.co.uk	ShopWiki	Whsmith
Inline Skates	amazon.co.uk	Google Products	decathlon.co.uk
Zip Wire Experience	buyagift.co.uk	Google Products	
Personalised Map	etsy.com	Google Products	
Waterproof Jacket	outdoorgear.co.uk	Google Products	Cotswold Outdoor
Sleeping Bag	Cotswold Outdoor	ShopWiki	Cotswold Outdoor
Day Out Ticket(s)	redletterdays.co.uk	ShopWiki	Debenhams
Helicopter Flight	buyagift.co.uk	ShopWiki	Whsmith
Paddle Boarding for Two	redletterdays.co.uk	Shopping.com	
Standup Comedy Tickets	eventim.co.uk	Google Products	Debenhams
Binoculars	amazon.co.uk	Kelkoo	Argos

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Black Friday

Advent calendar

gift items

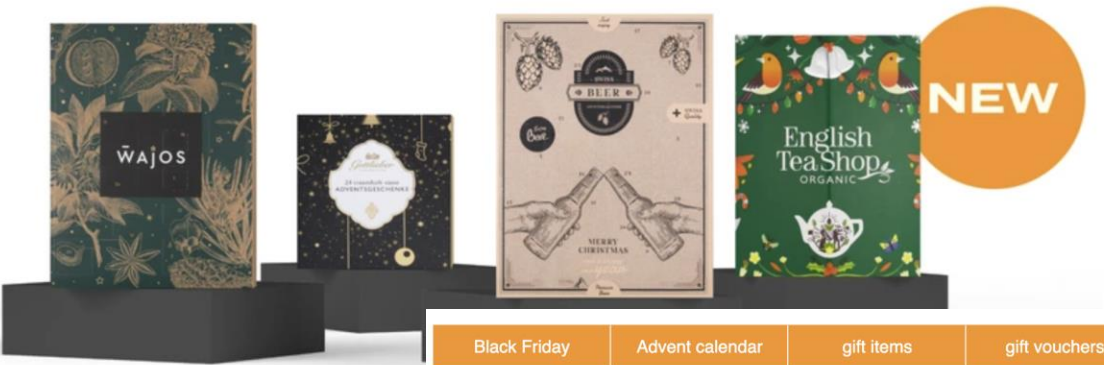
gift vouchers

gift sets

personalization

Recipient

#CHOOSE
YOUR
SURPRISE
SURPRISE



Black Friday

Advent calendar

gift items

gift vouchers

gift sets

personalization

Recipient

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[home page](#) · [gift items](#) · [gift sets](#)

Price CHF

CHF 8

CHF 449

gift sets

1-40 of 95 items

< 1 2 3 >

Most Popular

▼

category

- truffles (9)
- Mediterranean (25)
- Gourmet (13)
- BBQ (3)
- tea (3)
- Wellness (19)
- Baby (9)
- Jewelry (3)
- Garden & Outdoor (5)
- Wine & Beer (11)

browsing

 new products

 top seller



★★★★★

Premium Gift Box Gourmet
Personalizable

CHF 99.95



Premium Gift Set Pasta &
Lemon

CHF 39.95



Premium Gift Set Tea
Enjoyment & Honey "Time for
You"

CHF 59.95



Premium Gift Set Italia Large

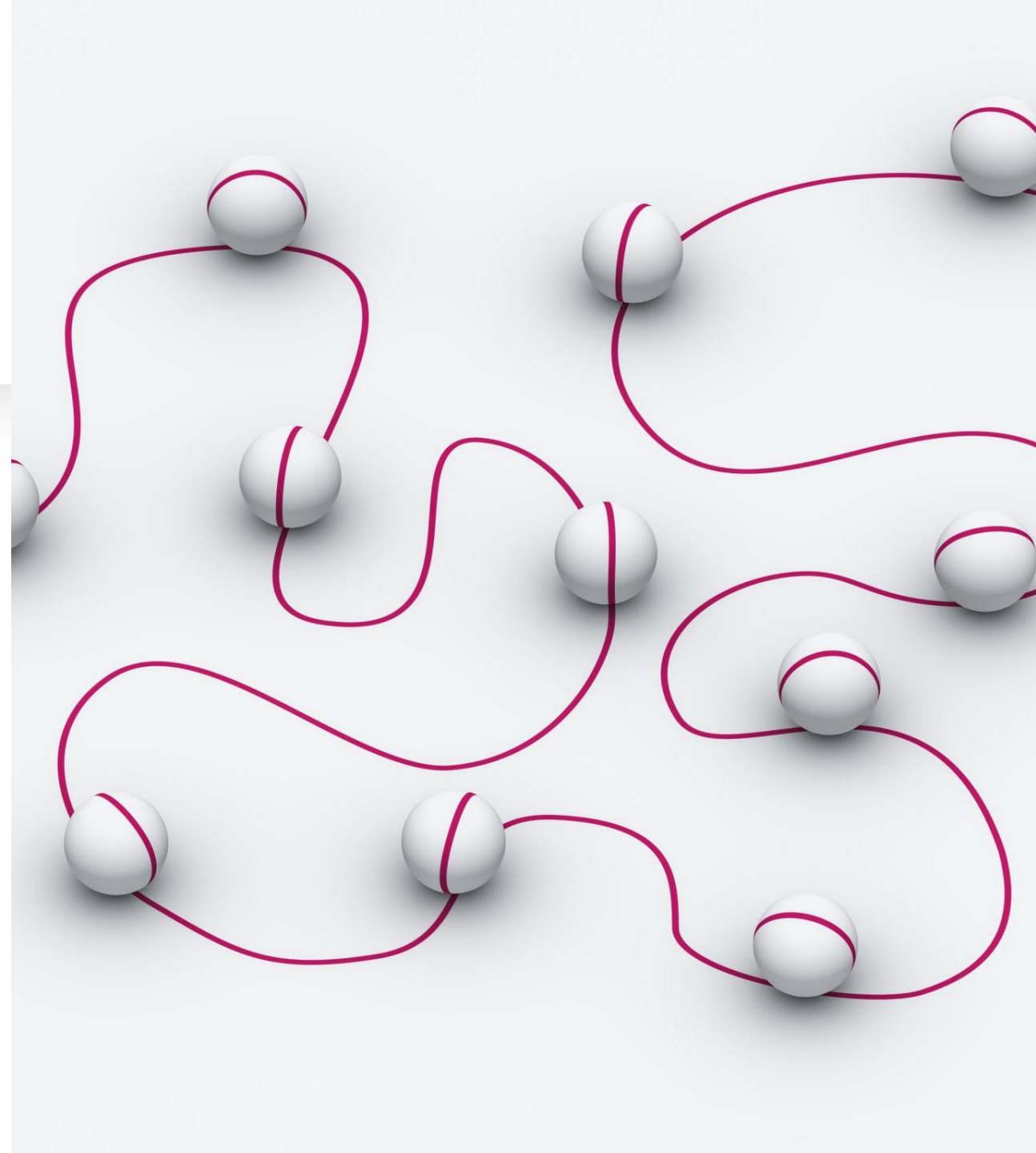
CHF 99.95 CHF 69.95



Geschenkidee.CH

CONCLUSION

The Fuzzy Giftanator blends the traditional gifting roots with advanced fuzzy logic to redefine modern gift-giving. By simplifying the decision-making process and integrating gamification, the app ensures a more personalized and enjoyable user experience. It transforms a challenging task into an exciting, human-centered journey.



REFERENCES

- <https://www.geschenkidee.ch/>
- <https://www.giftgen.co.uk/index-java.php?&TZ=1&ScrW=1536&ScrH=864&LNG=en-US&referrer=google.com>
- <https://giftlist.com/genie>

Our GitHub repo: <https://github.com/Vaibhav150123045/giftanator>

ANY QUESTIONS? 🤪

